

Chapter 4

Vanishing Theorems

Revised chapter corresponding to original pp. 221–267.

Original numbering and proof order retained.

The accompanying correction report gives the original page locations.

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In this chapter, we are still working with varieties over the complex numbers, and unless otherwise stated, all varieties are assumed to be non-singular. Schemes are of finite type over $\mathbb{C}$.

4.1 Preliminaries

If the cohomology of a sheaf F comes from topological cohomology, then there is a Kodaira-type vanishing theorem.

János Kollár

4.1A Simple Normal Crossings and Resolution of Singularities

Let X be a non-singular variety of dimension n .

Definition 4.1.1. An effective divisor $D = \sum D_i$ on X has *simple normal crossings* if D is reduced, each component D_i is smooth, and D is defined in a neighborhood of any point by an equation in local analytic coordinates of the type

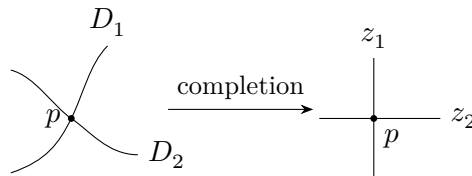
$$z_1 \cdots z_k = 0,$$

for some $k \leq n$. A $\mathbb{Q}$ -divisor $E = \sum a_i D_i$ has *simple normal crossing support* if its underlying reduced support $\sum_{a_i \neq 0} D_i$ has simple normal crossings.

Remark 4.1.2. In the above case we say that D is a simple normal crossing divisor, or an snc divisor. Local analytic coordinates are coordinates in the analytic local ring $\mathbb{C}\{z_1, \dots, z_n\}$. We can also describe this condition using formal coordinates in the completion

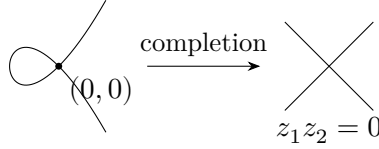
$$\widehat{\mathcal{O}}_{X,p} \cong \mathbb{C}[[z_1, \dots, z_n]],$$

where the isomorphism follows from the Cohen structure theorem; see [Eis95, 10.16]. The completion is a formal, rather than an analytic, local ring. For smooth components, the snc condition is equivalently that their local equations extend to a regular system of parameters. The number $k \leq n$ is determined by the components of D containing p : if $k = 0$, then $p \notin \text{Supp } D$; if $k \geq 1$, exactly k components contain p .



Remark 4.1.3. In our definition we emphasize that each component must be smooth; this is why we call it “simple” normal crossing. Some references on

complex manifolds allow locally smooth branches without requiring the global irreducible components to be smooth, and call this normal crossing. For example, in the affine plane $\mathbb{A}^2$ with coordinates x, y , the nodal curve $y^2 = x^2(x + 1)$ is irreducible but has a node at the origin. Therefore it is a normal crossing divisor on the complex manifold, but not a simple normal crossing divisor.



In other words, intersections among the components of an snc divisor should occur “as transversely as possible”: locally they look like distinct coordinate hyperplanes.

Example 4.1.4 (Hyperplane arrangements). Let $D = \sum H_i$ be the reduced divisor on $\mathbb{P}^n$ whose components are a collection of hyperplanes $H_i \subseteq \mathbb{P}^n$, corresponding to a finite set of points $S \subseteq (\mathbb{P}^n)^\vee$. Then D is snc if and only if the points of S are in linearly general position, i.e. for $r < n$, no $r + 2$ points lie on an r -dimensional linear subspace of $(\mathbb{P}^n)^\vee$.

Proof. By the rank–nullity theorem, the hyperplanes of $\mathbb{P}^n$ correspond to the points of its dual space.

“ $\Rightarrow$ ”, by contraposition: suppose S is not in linearly general position, so there are $r + 2$ points lying on an r -dimensional linear subspace, with $r < n$. Their linear forms

$$L_1, \dots, L_{r+2} \in \text{Hom}(\mathbb{C}^{n+1}, \mathbb{C})$$

span a subspace of dimension at most $r + 1$, and hence satisfy

$$a_1 L_1 + \dots + a_{r+2} L_{r+2} = 0$$

with the a_i not all zero. The codimension of $Y = \bigcap_{i=1}^{r+2} H_i$ equals the dimension of their span, which is at most $r + 1 \leq n$. Thus Y is non-empty. At a point $p \in Y$, dehomogenizing by a linear form nonzero at p gives dependent local equations. They cannot be part of a regular system of parameters in $\mathcal{O}_{\mathbb{P}^n, p}$, or of formal coordinates in $\widehat{\mathcal{O}}_{\mathbb{P}^n, p}$. Thus D is not snc.

“ $\Leftarrow$ ”, again by contraposition: suppose D is not snc at p . The equations $L_1, \dots, L_k$ of the hyperplanes containing p are linearly dependent. Write $d = \dim \text{Span}\{L_1, \dots, L_k\} < k$. Since their common kernel contains a representative of p , we have $d \leq n$. The corresponding points lie in a projective space of dimension $r = d - 1 < n$, and $k \geq d + 1 = r + 2$. Thus S is not in linearly general position. $\square$

The normal crossing condition is very important because it singles out a class of divisors whose singularities we can understand completely.

Theorem 4.1.5 (Resolution of singularities). *Let (X, D) be a pair where X is a complex algebraic variety, possibly singular, and $D \subseteq X$ is an effective Cartier divisor.*

- (i) *There is a projective birational morphism $\mu : X' \rightarrow X$, with X' non-singular and with divisorial exceptional locus $\text{Exc}(\mu)$, such that the reduced divisor supported on*

$$\text{Supp}(\mu^*D) \cup \text{Exc}(\mu)$$

is snc.

- (ii) *We can construct X' via a sequence of blow-ups along smooth centers in the successive bad loci of the pair. In particular, μ can be chosen to be an isomorphism over*

$$X \setminus (\text{Sing}(X) \cup \text{Sing}(D)).$$

We call X' a log resolution of D , or of the pair (X, D) . Log resolutions also exist over arbitrary fields of characteristic zero by Hironaka's theorem. Notice that μ^*D itself may have multiplicities; the assertion is about its support together with the exceptional locus.

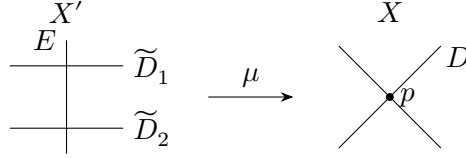


Figure 4.1: A schematic log resolution. The reduced total transform, including the exceptional divisors, is snc.

Corollary 4.1.6. *Assume X is smooth, and that $\mu : X' \rightarrow X$ is a log resolution of (X, D) constructed by a sequence of blow-ups along smooth centers.*

- (i) *For the canonical divisors of X and X' , we have*

$$\mu_*\mathcal{O}_{X'}(K_{X'}) = \mathcal{O}_X(K_X), \quad R^i\mu_*\mathcal{O}_{X'}(K_{X'}) = 0 \quad (i > 0).$$

- (ii) *If X is projective with an ample divisor H , then for suitable integers $p \gg 0$ and $b_j > 0$,*

$$\mu^*(pH) - \sum_j b_j E_j$$

is ample on X' , where the sum is over the μ -exceptional prime divisors.

Proof. We factor the resolution as

$$X' = X_m \xrightarrow{\pi_m} X_{m-1} \rightarrow \cdots \rightarrow X_1 \xrightarrow{\pi_1} X_0 = X,$$

where π_k blows up a smooth center Z_{k-1} of codimension $c_k \geq 2$. Blow-ups of Cartier centers can be omitted. Treating connected components of the centers separately, we may take each exceptional divisor E_k to be prime. Write $\mu_k = \pi_1 \circ \cdots \circ \pi_k$.

(i) First consider a single blow-up. By [Har77, II, Exercise 8.5],

$$K_{X_1} = \pi_1^* K_X + (c_1 - 1)E_1.$$

The projection formula gives

$$R^i \pi_{1*} \mathcal{O}_{X_1}(K_{X_1}) \cong R^i \pi_{1*} \mathcal{O}_{X_1}((c_1 - 1)E_1) \otimes \mathcal{O}_X(K_X).$$

Thus put $\mathcal{F} = \mathcal{O}_{X_1}((c_1 - 1)E_1)$.

- For $i = 0$, algebraic Hartogs and $\text{codim}_X Z_0 \geq 2$ give $\pi_{1*} \mathcal{F} = \mathcal{O}_X$.
- For $i > 0$, each fiber over Z_0 is $\mathbb{P}^{c_1-1}$ and

$$\mathcal{F}|_{\pi_1^{-1}(x)} \cong \mathcal{O}_{\mathbb{P}^{c_1-1}}(-(c_1 - 1)).$$

Its higher cohomology vanishes. However, π_1 is not flat, so we still need an argument with infinitesimal neighborhoods.

Work over an affine open $U \subseteq X$, and let I be the ideal of $Z_0 \cap U$. Since $I\mathcal{O}_{\pi_1^{-1}(U)} = \mathcal{O}(-E_1)$, the pullback of the thickened center $Z_k = V(I^k)$ is $F_k = kE_1$. Denote $\mathcal{F}_k = \mathcal{F}|_{F_k}$. Thus we have the Cartesian diagram

$$\begin{array}{ccc} F_k & \longrightarrow & \pi_1^{-1}(U) \\ \downarrow & & \downarrow \pi_1 \\ Z_k & \longrightarrow & U. \end{array}$$

These are thickenings of the exceptional divisor over the center. Because E_1 is Cartier, the actual short exact sequence is

$$0 \longrightarrow \mathcal{O}_{E_1}(-kE_1) \longrightarrow \mathcal{O}_{F_{k+1}} \longrightarrow \mathcal{O}_{F_k} \longrightarrow 0.$$

Writing $\mathcal{O}_{E_1}(1) = \mathcal{O}_{X_1}(-E_1)|_{E_1}$ and twisting by $\mathcal{F}$ gives

$$0 \longrightarrow \mathcal{O}_{E_1}(k - c_1 + 1) \longrightarrow \mathcal{F}_{k+1} \longrightarrow \mathcal{F}_k \longrightarrow 0.$$

The map $E_1 \rightarrow Z_0$ is a projective space bundle of relative dimension $c_1 - 1$. The projective space cohomology calculation gives

$$R^i(E_1 \rightarrow Z_0)_* \mathcal{O}_{E_1}(a) = 0 \quad \text{for } i > 0, a \geq -(c_1 - 1).$$

Since $Z_0 \cap U$ is affine, this also gives the corresponding vanishing of higher cohomology on $E_1 \cap \pi_1^{-1}(U)$. Starting with $\mathcal{F}_1 = \mathcal{O}_{E_1}(-(c_1 - 1))$, the long exact sequence therefore proves, by induction, that

$$H^i(F_k, \mathcal{F}_k) = 0 \quad (i > 0, k \geq 1).$$

The theorem of formal functions, applied to the ideal I , now gives

$$H^0(U, \widehat{R^i \pi_{1*} \mathcal{F}})^I \cong \varprojlim_k H^i(F_k, \mathcal{F}_k) = 0.$$

The sheaf $R^i \pi_{1*} \mathcal{F}$ is coherent and supported on Z_0 , because π_1 is an isomorphism outside Z_0 . Its vanishing completion implies its vanishing: locally at a point of Z_0 this follows from Nakayama's lemma, or from faithful flatness of completion. Hence $R^i \pi_{1*} \mathcal{F} = 0$ for $i > 0$.

Now suppose the assertion holds for k blow-ups. The Grothendieck spectral sequence for $\mu_{k+1} = \mu_k \circ \pi_{k+1}$ is

$$E_2^{p,q} = R^p \mu_{k*} (R^q \pi_{k+1*} \mathcal{O}_{X_{k+1}}(K_{X_{k+1}})) \implies R^{p+q} \mu_{k+1*} \mathcal{O}_{X_{k+1}}(K_{X_{k+1}}).$$

The single blow-up case shows that it collapses to the p -axis, so

$$R^n \mu_{k+1*} \mathcal{O}_{X_{k+1}}(K_{X_{k+1}}) \cong R^n \mu_{k*} \mathcal{O}_{X_k}(K_{X_k}).$$

The right side is $\mathcal{O}_X(K_X)$ when $n = 0$ and is zero when $n > 0$.

(ii) We again use induction. For a single blow-up, $\mathcal{O}_{X_1}(-E_1) = \mathcal{O}_{X_1}(1)$ is π_1 -very ample. Tensoring it with the pullback of a sufficiently high power of an ample line bundle on X makes it very ample on X_1 . Thus

$$A_1 = \pi_1^*(p_1 H) - E_1$$

is very ample for some $p_1 > 0$. Suppose that on X_k we have

$$A_k = \mu_k^*(p_k H) - \sum_{i=1}^k c_i^{(k)} E_i^{(k)}, \quad c_i^{(k)} > 0,$$

where $E_i^{(k)}$ is the strict transform of E_i . For a suitable $p'_{k+1} > 0$,

$$A_{k+1} = \pi_{k+1}^*(p'_{k+1} A_k) - E_{k+1}$$

is very ample. Since

$$\pi_{k+1}^* E_i^{(k)} = E_i^{(k+1)} + m_{k+1,i} E_{k+1}, \quad m_{k+1,i} \geq 0,$$

we obtain

$$\begin{aligned} A_{k+1} &= \mu_{k+1}^*(p'_{k+1} p_k H) - \sum_{i=1}^k p'_{k+1} c_i^{(k)} E_i^{(k+1)} \\ &\quad - \left(1 + \sum_{i=1}^k p'_{k+1} c_i^{(k)} m_{k+1,i} \right) E_{k+1} \\ &= \mu_{k+1}^*(p_{k+1} H) - \sum_{i=1}^{k+1} c_i^{(k+1)} E_i^{(k+1)}. \end{aligned}$$

All these coefficients are positive, as required. $\square$

Remark 4.1.7 (Tangent description of simple normal crossings). The coordinate definition says that the smooth components intersect transversally everywhere. We can also describe this by a regular system of parameters, i.e. minimal generators of $\mathfrak{m}_p$ whose classes form a basis of the Zariski cotangent space $\mathfrak{m}_p/\mathfrak{m}_p^2$.

At a closed point p , if the components through p have local equations $f_1, \dots, f_k$, the condition is that these extend to parameters $f_1, \dots, f_n$ of $\mathcal{O}_{X,p}$. Then

$$D : f_1 \cdots f_k = 0, \quad \mathfrak{m}_p/\mathfrak{m}_p^2 = \text{Span}_{\mathbb{C}}\{\bar{f}_1, \dots, \bar{f}_n\}.$$

For $R = \mathcal{O}_{X,p}$, the natural isomorphism $\text{gr}_{\mathfrak{m}} R \cong \text{gr}_{\widehat{\mathfrak{m}}} \widehat{R}$ identifies the initial classes. More precisely, the chosen parameters give the isomorphism

$$\mathbb{C}[[z_1, \dots, z_n]] \xrightarrow{\sim} \widehat{R}, \quad z_i \mapsto f_i.$$

Analytically, their independent differentials give local coordinates by the inverse function theorem. Thus these descriptions agree.

4.1B Covering Trick

We introduce an exquisite method to extract “roots” of divisors or line bundles.

Cyclic coverings.

We begin with the local description of the m -fold cyclic covering branched along a given divisor. Suppose X is affine, with $H^0(X, \mathcal{O}_X) = \mathbb{C}[X]$, and $s \in \mathbb{C}[X]$ is nonzero. We wish to construct a scheme Y on which the m -th root of s makes sense. In $X \times \mathbb{A}^1$, with coordinate t on the second factor, set

$$Y = V(t^m - s), \quad \mathbb{C}[Y] = \mathbb{C}[X][t]/(t^m - s).$$

Projection gives a finite flat morphism $\pi : Y \rightarrow X$ of degree m , branched over $D = \text{div}(s)$ when $m > 1$. The function $s' = t|_Y$ satisfies

$$(s')^m = \pi^* s.$$

Thus we have extracted the desired root. The scheme Y need not be irreducible. This local construction can be globalized as follows.

Proposition 4.1.8 (Cyclic coverings). *Let X be a variety, and L a line bundle on X . Suppose a positive integer $m \geq 1$ and a nonzero section $s \in H^0(X, L^{\otimes m})$ are given, defining an effective Cartier divisor $D = \text{div}(s)$. Then there is a finite flat covering*

$$\pi : Y \longrightarrow X$$

of degree m , where Y is a scheme, such that $L' = \pi^ L$ carries a section*

$$s' \in H^0(Y, L'), \quad (s')^m = \pi^* s.$$

The Cartier divisor $D' = \text{div}(s')$ maps isomorphically to D . Moreover, if X and D are non-singular, then so are Y and D' .

As above, we can regard s' as the m -th root of s .

Proof. Locally we trivialize L on affine open sets and carry out the preceding construction. To see directly that these local constructions glue, consider the geometric line bundle

$$\mathbf{L} = \mathbf{V}(L^\vee) = \operatorname{Spec}_X \operatorname{Sym}(L^\vee), \quad p : \mathbf{L} \longrightarrow X.$$

Since $p_* \mathcal{O}_{\mathbf{L}} = \operatorname{Sym}(L^\vee)$, a section of $p^* L$ corresponds to a module homomorphism

$$\begin{aligned} H^0(\mathbf{L}, p^* L) &= \operatorname{Hom}_{\mathcal{O}_{\mathbf{L}}}(\mathcal{O}_{\mathbf{L}}, p^* L) \\ &\cong \operatorname{Hom}_{p_* \mathcal{O}_{\mathbf{L}}}(p_* \mathcal{O}_{\mathbf{L}}, p_* p^* L) \\ &\cong \operatorname{Hom}_{\operatorname{Sym}(L^\vee)}(\operatorname{Sym}(L^\vee), L \otimes \operatorname{Sym}(L^\vee)). \end{aligned}$$

Here we use the equivalence for an affine morphism and the projection formula. In rank one, symmetric powers coincide with tensor powers, and

$$\begin{aligned} L \otimes \operatorname{Sym}(L^\vee) &= \bigoplus_{n \geq 0} L \otimes (L^\vee)^{\otimes n} \\ &\cong L \oplus \mathcal{O}_X \oplus L^\vee \oplus (L^\vee)^{\otimes 2} \oplus \dots \end{aligned}$$

The inclusion of $\operatorname{Sym}(L^\vee)$ into the second and subsequent summands, with this degree shift, is a $\operatorname{Sym}(L^\vee)$ -module map. It defines the tautological section

$$T \in H^0(\mathbf{L}, p^* L).$$

Take $Y \subseteq \mathbf{L}$ to be the zero scheme of

$$T^m - p^* s \in H^0(\mathbf{L}, p^* L^{\otimes m}),$$

and put $s' = T|_Y$. For $\pi = p|_Y$, we immediately have $(s')^m = \pi^* s$.

Let $U = \operatorname{Spec} A$ trivialize L , with $H^0(U, L) = Ae$, and let t be the dual basis. Write $s|_U = fe^{\otimes m}$. Then

$$\mathbf{L}|_U = \operatorname{Spec} A[t] = U \times \mathbb{A}^1, \quad T = tp^* e,$$

so Y has equation $t^m - f = 0$ over U . The divisors D and D' have equations $f = 0$ and $t = 0$, respectively, and

$$(A[t]/(t^m - f))/(t) \cong A/(f).$$

Also, t is a nonzerodivisor in $A[t]/(t^m - f)$: if $tb = 0$, then $fb = t^m b = 0$, and f is a nonzerodivisor on this free A -module. Thus D' is Cartier and maps isomorphically to D . Finally,

$$A[t]/(t^m - f) = A \oplus At \oplus \dots \oplus At^{m-1}$$

is free of rank m , proving finiteness, flatness and surjectivity.

For smoothness, if $m = 1$ the cover is the graph of f . If $m > 1$, the differential of $F = t^m - f$ is

$$dF = mt^{m-1} dt - df.$$

At a point away from D' , the coefficient of dt is nonzero. At a point of D' , smoothness of D gives $df \neq 0$. Since the dt direction is independent of the directions on X , dF never vanishes on Y . Thus Y is non-singular, and so is $D' \cong D$. $\square$

Remark 4.1.9 (Geometrical grasp). Restrict to $U = \operatorname{Spec} A$, where $H^0(U, L) = Ae$. Then $\mathbf{L}|_U = \operatorname{Spec} A[t]$, where t is dual to e , and by the definition of pullback,

$$H^0(\mathbf{L}|_U, p^*L) = Ae \otimes_A A[t].$$

This is the restriction of $L \otimes \operatorname{Sym}(L^\vee)$ after sheafification. The tautological section comes from the module map

$$\varphi : A[t] \longrightarrow Ae \otimes_A A[t], \quad 1 \mapsto e \otimes t.$$

The pullback of the local section e is

$$e \mapsto p^*e = e \otimes 1,$$

so $T = tp^*e$. For a closed point $a \in \mathbf{L}|_U$ with $x = p(a)$, we have

$$\begin{aligned} \mathbf{L}_x &= \mathbf{L} \times_X \operatorname{Spec} k(x) = \operatorname{Spec}(A[t] \otimes_A \mathbb{C}) = \mathbb{A}_{\mathbb{C}}^1 \\ &= \operatorname{Spec} \operatorname{Sym}_{\mathbb{C}}((Ae \otimes_A \mathbb{C})^\vee). \end{aligned}$$

From the functorial point of view,

$$\begin{aligned} \mathbf{L}_x(\mathbb{C}) &= \operatorname{Hom}_{\mathbb{C}\text{-schemes}}(\operatorname{Spec} \mathbb{C}, \mathbf{L}_x) \\ &= \operatorname{Hom}_{\mathbb{C}\text{-algebras}}(\operatorname{Sym}_{\mathbb{C}}((Ae \otimes_A \mathbb{C})^\vee), \mathbb{C}) \\ &\cong \operatorname{Hom}_{\mathbb{C}}((Ae \otimes_A \mathbb{C})^\vee, \mathbb{C}) \\ &\cong Ae \otimes_A \mathbb{C} = \mathbb{C}e. \end{aligned}$$

Thus we can write $a = \lambda e$, with $\lambda \in \mathbb{C}$, and

$$T(a) = (tp^*e)(a) = t(a)(e \otimes 1) = \lambda e = a,$$

under the identification $(p^*L)_a = L_x$. This is why it is called the “tautological” section: at a point of the geometric line bundle it returns that vector itself.

Remark 4.1.10. From the construction,

$$p_*\mathcal{O}_{\mathbf{L}} = \operatorname{Sym}(L^\vee) = \mathcal{O}_X \oplus L^{-1} \oplus L^{-2} \oplus \dots.$$

Locally this is $A[t] = A \oplus At \oplus At^2 \oplus \dots$. Passing to the quotient defining Y , the relation $t^m = f$ gives

$$A[t]/(t^m - f) = A \oplus At \oplus \dots \oplus At^{m-1}.$$

These descriptions glue to the canonical isomorphism of $\mathcal{O}_X$ -modules

$$\pi_* \mathcal{O}_Y = \mathcal{O}_X \oplus L^{-1} \oplus L^{-2} \oplus \dots \oplus L^{1-m}.$$

The multiplication is determined by s . Moreover, $D' \sim_{\text{lin}} \pi^* L$ because D' is the divisor of the section s' of $\pi^* L$.

Remark 4.1.11. This cyclic covering construction is all that is needed for the classical Kodaira vanishing theorem in the next section.

Proposition 4.1.12 (An snc divisor behaves well). *In the above situation, suppose X and D are smooth, and $D + \sum_{i=2}^r D_i$ is a reduced snc divisor on X . Put $D'_i = \pi^* D_i$ for $i \geq 2$. Then*

$$mD' + \sum_{i=2}^r D'_i$$

has simple normal crossing support on Y ; each D'_i is smooth and reduced.

Proof. We use the tangent description in Remark 4.1.7. Let $q \in Y$, and put $p = \pi(q)$. If $p \notin D$, the covering is *étale* near q , so the snc condition is preserved.

If $p \in D$, choose regular parameters $f, f_2, \dots, f_n$ at p , where D has equation $f = 0$ and the other components through p have equations among the f_i . Then

$$\mathcal{O}_{Y,q} = \mathcal{O}_{X,p}[t]/(t^m - f).$$

To justify that this ring is already local, note that it is finite over the local ring $\mathcal{O}_{X,p}$, so every maximal ideal contracts to $\mathfrak{m}_p$. Its quotient by $\mathfrak{m}_p$ is $\mathbb{C}[t]/(t^m)$, which has only the maximal ideal (t) . Thus its unique maximal ideal is

$$\mathfrak{m}_q = (t, f, f_2, \dots, f_n) = (t, f_2, \dots, f_n).$$

These n elements form a regular system of parameters, since Y is smooth of dimension n . The relevant divisors are $t = 0$ and some of the $f_i = 0$, so their reduced union is snc. $\square$

Bloch–Gieseker and Kawamata coverings.

Lemma 4.1.13. *On an arbitrary quasi-projective variety X , every line bundle L can be written*

$$L = A \otimes B^\vee,$$

where A and B are very ample line bundles.

Proof. Choose an immersion $X \hookrightarrow Z \hookrightarrow \mathbb{P}^N$, with X open in the projective variety Z , and let $H = \mathcal{O}_Z(1)|_X$. This is very ample and ample. Indeed, every coherent sheaf F on X extends to a coherent sheaf F' on Z , and Serre's theorem makes $F' \otimes \mathcal{O}_Z(n)$ globally generated for $n \gg 0$. Its restriction $F \otimes H^n$ is therefore globally generated.

Choose n_1 such that $L \otimes H^{n_1}$ is globally generated. Its tensor product with the very ample line bundle H is very ample. Hence, for a suitable $n > 0$, we can take

$$A = L \otimes H^n, \quad B = H^n.$$

□

Theorem 4.1.14 (BG coverings). *Let X be a quasi-projective variety of dimension n , let L be a line bundle on X , and fix a positive integer m . There exist a variety Y , a finite flat surjective morphism $f : Y \rightarrow X$, and a line bundle N on Y such that $f^*L \cong N^{\otimes m}$.*

*If X is non-singular, Y can be chosen non-singular. In this case, given an snc divisor $D = \sum D_i$ on X , we can arrange that each f^*D_i is smooth and f^*D is reduced and snc. Finally, if $\dim X \geq 2$, each f^*D_i can also be chosen irreducible.*

Proof. Step 1: We first make some reductions. If $\dim X = 0$, then $X = \text{Spec } \mathbb{C}$ and L is trivial, so the identity suffices. Assume $n \geq 1$. Suppose the result is known for the pullback of the hyperplane bundle by a morphism $X \rightarrow \mathbb{P}^r$ with finite fibers. By the preceding lemma, write $L = A \otimes B^\vee$ for two very ample line bundles. First take a covering $f_1 : Y_1 \rightarrow X$ with $f_1^*A = N_A^{\otimes m}$. Apply the special case again to f_1^*B : its map to projective space still has finite fibers. This gives $f_2 : Y \rightarrow Y_1$ with $f_2^*f_1^*B = N_B^{\otimes m}$. Then, for $f = f_1 \circ f_2$,

$$\begin{aligned} f^*L &= f_2^*(f_1^*A) \otimes f_2^*(f_1^*B^\vee) \\ &\cong f_2^*N_A^{\otimes m} \otimes N_B^{-m} = (f_2^*N_A \otimes N_B^\vee)^{\otimes m}. \end{aligned}$$

The extra smoothness and snc requirements can be retained at both stages.

Step 2: Thus assume $L = \varphi^*\mathcal{O}_{\mathbb{P}^r}(1)$ for a finite-to-one morphism $\varphi : X \rightarrow \mathbb{P}^r$. Fix the Fermat covering

$$\nu : \mathbb{P}^r \longrightarrow \mathbb{P}^r, \quad [x_0 : \cdots : x_r] \longmapsto [x_0^m : \cdots : x_r^m].$$

It satisfies $\nu^*\mathcal{O}(1) = \mathcal{O}(m)$. On standard affine charts its algebra is free with basis the monomials whose exponents are between 0 and $m-1$, so ν is finite flat of degree m^r .

For $g \in G = \text{GL}(r+1, \mathbb{C})$, let $\nu_g = g \circ \nu$ and set

$$Y_g = X \times_{\mathbb{P}^r} \mathbb{P}^r, \quad \begin{array}{ccc} Y_g & \xrightarrow{\bar{\varphi}} & \mathbb{P}^r \\ f_g \downarrow & & \downarrow \nu_g \\ X & \xrightarrow{\varphi} & \mathbb{P}^r. \end{array}$$

Here the fiber product uses φ and ν_g . The morphism f_g is finite flat and surjective by base change, and

$$f_g^*L = \bar{\varphi}^*\nu_g^*\mathcal{O}(1) = \bar{\varphi}^*\mathcal{O}(m) = (\bar{\varphi}^*\mathcal{O}(1))^{\otimes m}.$$

We show that a general g gives the remaining properties. The space Y_g is the inverse image of the graph of g^{-1} under

$$\varphi \times \nu : X \times \mathbb{P}^r \longrightarrow \mathbb{P}^r \times \mathbb{P}^r.$$

The image has dimension $n + r > r$, since φ has finite fibers. The generic irreducibility result [Laz04a, Example 3.3.10] therefore shows that Y_g is irreducible for general g .

For reducedness, choose g so that none of the translated coordinate hyperplanes contains $\varphi(X)$. Over a dense open subset of X , the covering is then *étale*, since the Fermat covering is *étale* off these hyperplanes. This gives a reduced generic fiber. Finite flatness over the integral variety X then gives reducedness everywhere by the following lemma. $\square$

Lemma 4.1.15. *Let $f : V \rightarrow W$ be a finite flat morphism of schemes of finite type over $\mathbb{C}$, with W integral and V generically reduced. Then V is reduced.*

Proof. Locally write $V = \text{Spec } B$ and $W = \text{Spec } A$, where A is a domain and B is finite flat over A . Flatness implies that every irreducible component of V dominates W , and that

$$B \longrightarrow B \otimes_A \text{Frac}(A)$$

is injective. The generic fiber is a zero-dimensional Noetherian scheme; its local rings are the reduced local rings at the generic points of V . Hence it is reduced. A nilpotent element of B maps to zero in this generic fiber and is therefore zero. Equivalently, after some localization B_δ is reduced, and $\delta^a b = 0$ implies $b = 0$ by flatness. $\square$

Continuation of the proof of Theorem 4.1.14. The same argument, applied to a reduced irreducible Cartier divisor $E \subseteq X$, makes f_g^*E reduced and irreducible if $\dim X \geq 2$, so that $\dim E \geq 1$.

Step 3: Suppose X is non-singular. Consider

$$\begin{array}{ccccc} W = X \times_{\mathbb{P}^r} (G \times \mathbb{P}^r) & \longrightarrow & G \times \mathbb{P}^r & \xrightarrow{\text{pr}_1} & G \\ n \downarrow & & \downarrow m & & \\ X & \xrightarrow{\varphi} & \mathbb{P}^r, & & \end{array}$$

where $m(g, y) = g \cdot \nu(y)$ and $b : W \rightarrow G$ denotes the top composite. Its fiber over g is Y_g . The transitive group action makes m smooth, and hence n is smooth by base change. Since X is smooth, so is W . Generic smoothness applied to b makes

Y_g smooth for general g . Applying the same argument to a smooth divisor $E \subseteq X$ makes f_g^*E smooth as well.

Step 4: Let $D = \sum D_i$ be snc. The subvarieties

$$F_i = n^{-1}(D_i) = D_i \times_{\mathbb{P}^r} (G \times \mathbb{P}^r) \subseteq W$$

are smooth and meet transversally. Indeed, local equations of the D_i form part of a regular system of parameters on X . Their pullbacks under the smooth map n remain part of a regular system of parameters on W : a smooth local map is flat with regular fibers, and one adds parameters along the fiber to those from the base.

The next lemma shows that, for general g , the fibers $(F_i)_g = f_g^*D_i$ are smooth and meet transversally in Y_g . Thus f_g^*D is snc. All the conditions imposed on g hold on finitely many dense open subsets, so they can be imposed simultaneously. $\square$

Lemma 4.1.16. *Let $b : W \rightarrow G$ be a morphism of non-singular varieties, and let $V_i \subseteq W$ be a finite collection of smooth subvarieties meeting transversally. For general $g \in G$, the fibers $(V_i)_g$ are smooth and meet transversally, or not at all, in W_g .*

Proof. Apply generic smoothness to W and to all nonempty intersections $V_I = \bigcap_{i \in I} V_i$. Remove the closures of the images of components that do not dominate G . There are only finitely many such strata, so one dense open subset of G works for all of them. Over it, each nonempty $(V_I)_g$ is smooth, with codimension in W_g equal to the codimension of V_I in W . These codimensions are the sums of those of the V_i , since the intersections in W are transverse. The same is therefore true in W_g , which gives transversality there. $\square$

By combining cyclic and BG coverings, we can pass to a covering on which each component of an snc divisor becomes divisible by a prescribed positive integer.

Proposition 4.1.17 (Kawamata coverings). *Let X be a non-singular quasi-projective variety, let $D = \sum_{i=1}^r D_i$ be snc, and fix positive integers $m_1, \dots, m_r$. There exist a smooth variety Y and a finite flat covering $h : Y \rightarrow X$ such that*

$$h^*D_i = m_i D'_i,$$

where each D'_i is smooth and $D' = \sum D'_i$ is snc.

Proof. By induction, it suffices to construct a cover with

$$h^*D_1 = m_1 D'_1, \quad h^*D_i = D'_i \quad (2 \leq i \leq r),$$

and $\sum D'_i$ snc. First take a BG covering $f : X_1 \rightarrow X$ such that

$$D_1^+ := f^*D_1 \sim_{\text{lin}} m_1 A_1^+,$$

for some divisor A_1^+ , while f^*D is snc and D_1^+ is smooth. Then take the m_1 -fold cyclic covering $\pi : Y \rightarrow X_1$ branched along D_1^+ . We have

$$\pi^*D_1^+ = m_1D'_1, \quad D'_1 \cong D_1^+.$$

Set $h = f \circ \pi$. For $i \geq 2$, put $D'_i = h^*D_i$. Proposition 4.1.12 says that $\sum D'_i$ is snc. The cyclic cover is integral: for $m_1 > 1$, its equation is Eisenstein at the generic point of any component of the nonempty reduced divisor D_1^+ ; its finite flat algebra embeds into the resulting function field. Thus Y is a smooth variety.

Replace X by Y and proceed with D'_2 , and so on. At every stage the previous root divisors have smooth reduced pullbacks. A smooth divisor may be disconnected, but its components are disjoint; the same construction applies to it. The multiplicities already extracted are therefore retained. $\square$

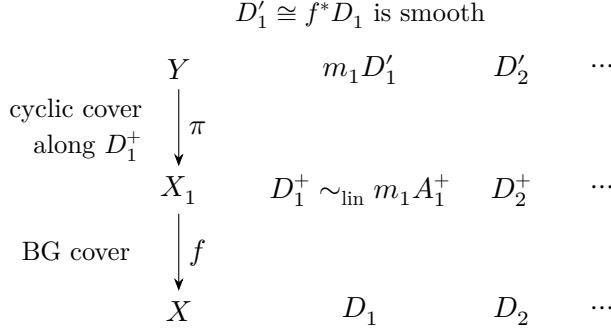


Figure 4.2: Inductive construction. Then replace X by Y and proceed with D'_2 .

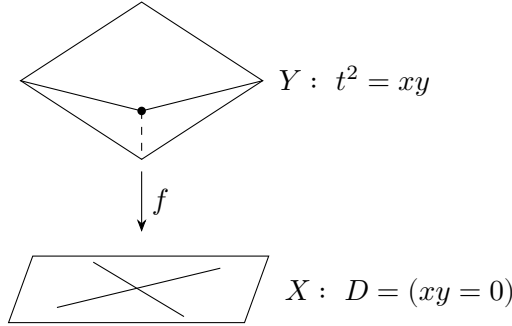


Figure 4.3: A schematic double cover branched along two crossing components. The local equation $t^2 = xy$ is singular at the origin. This differs from taking roots successively along smooth divisors as in Figure 4.2.

Injectivity.

We now use these covering constructions to establish vanishing theorems. Starting from a line bundle L on X , the general strategy is to pull it back to

a covering $f : Y \rightarrow X$ on which the problem simplifies. The next lemma shows that vanishing on Y is enough.

Lemma 4.1.18. *Let $f : Y \rightarrow X$ be a finite surjective morphism of complex algebraic varieties, with X normal, and let E be a vector bundle on X . Then the natural map*

$$H^i(X, E) \longrightarrow H^i(Y, f^*E)$$

*is injective. In particular, $H^i(Y, f^*E) = 0$ implies $H^i(X, E) = 0$.*

Proof. Finiteness and the projection formula give

$$H^i(Y, f^*E) \cong H^i(X, f_*f^*E) \cong H^i(X, E \otimes f_*\mathcal{O}_Y).$$

On the other hand, regard $\mathbb{C}(Y)$ as a finite-dimensional $\mathbb{C}(X)$ -vector space. For $b \in \mathbb{C}(Y)$, multiplication

$$\mathbb{C}(Y) \longrightarrow \mathbb{C}(Y), \quad x \longmapsto bx$$

is a linear transformation. Its matrix trace defines $\text{Tr}(b) \in \mathbb{C}(X)$. Locally, f is $\text{Spec } B \rightarrow \text{Spec } A$ with B finite, hence integral, over the normal domain A . For $b \in B$, its trace is integral over A and lies in $\text{Frac}(A)$, so it belongs to A . These local maps glue to

$$\text{Tr}_{Y/X} : f_*\mathcal{O}_Y \longrightarrow \mathcal{O}_X.$$

Write $d = [\mathbb{C}(Y) : \mathbb{C}(X)]$. The composite of the natural inclusion with the trace is multiplication by d . Thus the normalized trace

$$\frac{1}{d} \text{Tr}_{Y/X} : f_*\mathcal{O}_Y \longrightarrow \mathcal{O}_X$$

splits the natural inclusion $\mathcal{O}_X \hookrightarrow f_*\mathcal{O}_Y$. Locally this splits the exact sequence

$$0 \longrightarrow A \longrightarrow B \longrightarrow \text{coker} \longrightarrow 0.$$

Tensoring the splitting with E gives a splitting of $E \hookrightarrow E \otimes f_*\mathcal{O}_Y$. Hence $H^i(X, E)$ embeds as a direct summand of $H^i(X, E \otimes f_*\mathcal{O}_Y) \cong H^i(Y, f^*E)$, as required. $\square$

4.2 Kodaira and Nakano Vanishing Theorems

In this section we prove the classical vanishing theorems of Kodaira and Nakano. We first connect singular cohomology with the cohomology of the constant sheaf. On a locally contractible paracompact space, such as a smooth manifold, the comparison theorem gives

$$H_{\text{sing}}^i(X, G) \cong H^i(X, \underline{G}),$$

where G is an abelian group (in particular, the additive group of a commutative ring). For $G = \mathbb{C}$, the de Rham theorem further identifies these groups with the cohomology of the complex of smooth differential forms. Here and below, topological statements concern the analytification of X .

We also need the Lefschetz hyperplane theorem; see [Laz04a, 3.1.17], [Voi03, 1.23], or [GH78, p. 156].

Theorem 4.2.1 (Lefschetz hyperplane theorem). *Let X be a non-singular complex projective variety of dimension n , and let D be an effective ample divisor on X . The restriction*

$$r_i : H_{\text{sing}}^i(X, \mathbb{Z}) \longrightarrow H_{\text{sing}}^i(D, \mathbb{Z})$$

is an isomorphism for $i \leq n - 2$ and is injective for $i = n - 1$.

After changing coefficients to $\mathbb{C}$ and using the comparison above, we can use Hodge decomposition and the Dolbeault theorem; see [GH78] or [Voi02].

Theorem 4.2.2 (Hodge decomposition). *For a compact Kähler manifold X , the cohomology of the constant complex sheaf satisfies*

$$H^i(X, \mathbb{C}) = \bigoplus_{p+q=i} H^{p,q}(X), \quad H^{p,q}(X) = \overline{H^{q,p}(X)}.$$

Here $H^{p,q}(X) \subseteq H^{p+q}(X, \mathbb{C})$ consists of classes represented by closed forms of type (p, q) . Moreover, the Kähler identity $\Delta_d = 2\Delta_{\bar{\partial}}$ gives

$$H^{p,q}(X) \cong H_{\bar{\partial}}^{p,q}(X).$$

The Fubini–Study metric makes every smooth complex projective variety a compact Kähler manifold.

Theorem 4.2.3 (Dolbeault theorem). *For a complex manifold X , we have*

$$H^q(X, \Omega_X^p) \cong H_{\bar{\partial}}^{p,q}(X),$$

where Ω_X^p is the sheaf of holomorphic p -forms.

For a projective algebraic variety, GAGA identifies these coherent analytic cohomology groups with their algebraic counterparts. We use this identification without changing notation.

4.2A Kodaira Vanishing Theorem

Lemma 4.2.4. *Let X be a smooth complex projective variety of dimension n , and let $D \subseteq X$ be a smooth effective ample divisor. The restriction maps*

$$r_{p,q} : H^q(X, \Omega_X^p) \longrightarrow H^q(D, \Omega_D^p)$$

are bijective for $p + q \leq n - 2$ and injective for $p + q = n - 1$.

Proof. Combine the preceding three theorems. Restriction preserves the Hodge decomposition, so the isomorphism or injection on total cohomology restricts to an isomorphism or injection on each (p, q) summand. $\square$

We shall now turn to the central topic.

Theorem 4.2.5 (Kodaira vanishing). *Let X be a smooth complex projective variety of dimension n , and let A be an ample divisor on X . Then*

$$H^i(X, \mathcal{O}_X(K_X + A)) = 0 \quad (i > 0).$$

Equivalently, by Serre duality,

$$H^j(X, \mathcal{O}_X(-A)) = 0 \quad (j < n).$$

Proof. The case $n = 0$ is immediate. Since A is ample, Bertini's theorem gives a smooth divisor $D \in |mA|$ for $m \gg 0$. Let $\pi : Y \rightarrow X$ be the cyclic covering in Proposition 4.1.8, and put $A' = \pi^*A$. The tautological section gives a smooth divisor $D' \in |A'|$ with $\pi^*D = mD'$. It is ample because π is finite. By Lemma 4.1.18,

$$H^j(X, \mathcal{O}_X(-A)) \hookrightarrow H^j(Y, \pi^*\mathcal{O}_X(-A)) = H^j(Y, \mathcal{O}_Y(-D')).$$

Thus it suffices to prove vanishing for the negative of a smooth effective ample divisor.

Now let $D' \subseteq Y$ be such a divisor. Lemma 4.2.4 with $p = 0$ says that

$$H^j(Y, \mathcal{O}_Y) \rightarrow H^j(D', \mathcal{O}_{D'})$$

is bijective for $j \leq n - 2$ and injective for $j = n - 1$. The exact sequence

$$0 \rightarrow \mathcal{O}_Y(-D') \rightarrow \mathcal{O}_Y \rightarrow \mathcal{O}_{D'} \rightarrow 0$$

therefore gives $H^j(Y, \mathcal{O}_Y(-D')) = 0$ for $j \leq n - 1$: the map in degree $j - 1$ is surjective, and the map in degree j is injective. For $j = 0$, the negative-degree group is zero. This proves the theorem. $\square$

4.2B Differential Forms with Logarithmic Poles

The preceding proof uses only a small part of Lemma 4.2.4. The remaining instances lead to Nakano's generalization of Kodaira vanishing. Before giving the proof, we investigate differential forms with logarithmic poles; see [EV92].

For an snc divisor $D = \sum_{i=1}^r D_i$ on a non-singular n -fold X , let $\tau : U = X \setminus D \hookrightarrow X$ and write

$$\Omega_X^p(*D) = \varinjlim_{\nu} \Omega_X^p(\nu D) = \tau_* \Omega_U^p.$$

These forms may have poles of arbitrary order along D . Of course, $(\Omega_X^\bullet(*D), d)$ is a cochain complex.

Definition 4.2.6. The sheaf $\Omega_X^p(\log D)$ of differential forms with logarithmic poles along D is the subsheaf of $\Omega_X^p(*D)$ whose sections α satisfy the condition that both α and $d\alpha$ have at most simple poles along D .

Proposition 4.2.7 (Properties of differential log-forms). (a) $(\Omega_X^\bullet(\log D), d)$ is a subcomplex of $(\Omega_X^\bullet(*D), d)$.

(b) $\Omega_X^p(\log D) = \bigwedge^p \Omega_X^1(\log D)$.

(c) $\Omega_X^p(\log D)$ is locally free. More precisely, at a closed point x through which exactly $D_1, \dots, D_s$ pass, choose parameters $f_1, \dots, f_n$ with $D_i = (f_i = 0)$ for $i \leq s$. Set

$$\delta_i = \begin{cases} df_i/f_i & i \leq s, \\ df_i & i > s, \end{cases} \quad \delta_I = \delta_{i_1} \wedge \dots \wedge \delta_{i_p}$$

for $I = \{i_1 < \dots < i_p\}$. Then the δ_I , for $|I| = p$, form a free system of local generators.

In particular, $\Omega_X^0(\log D) = \mathcal{O}_X$.

Proof. (a) follows from $d^2 = 0$, and (b) follows from the generators in (c).

For (c), each δ_i is closed, since

$$d(df_i/f_i) = d(f_i^{-1}) \wedge df_i + f_i^{-1}d(df_i) = 0.$$

Hence the forms δ_I and the differentials of their regular linear combinations have at most simple poles. If Ω denotes their $\mathcal{O}_X$ -span, then $\Omega \subseteq \Omega_X^p(\log D)$. The generators are independent because they are nonzero scalar multiples of the usual basis df_I over the function field.

First consider a single component, with equation $f_1 = 0$. Write

$$\alpha = \alpha_1 + \alpha_2 \wedge \frac{df_1}{f_1},$$

where α_1 and α_2 involve only wedge products of $df_2, \dots, df_n$. If α and $d\alpha$ have at most simple poles, then

$$f_1\alpha = f_1\alpha_1 + \alpha_2 \wedge df_1, \quad f_1d\alpha = f_1d\alpha_1 + d\alpha_2 \wedge df_1$$

are regular. Comparing the terms with and without df_1 , we see that α_2 and $f_1\alpha_1$ are regular. Also,

$$d(f_1\alpha_1) = df_1 \wedge \alpha_1 + f_1d\alpha - d\alpha_2 \wedge df_1.$$

Thus $df_1 \wedge \alpha_1$ is regular, and, since α_1 does not involve df_1 , α_1 is regular. This proves the assertion for one component.

For several components, expand $\alpha = \sum h_I \delta_I$ with rational coefficients. At the generic point of each D_i , the other local equations are units, so the one-component argument shows that every h_I is regular there. Away from D they are already regular. A regular local ring is normal, and a rational function regular at all its height-one primes is regular. Thus every h_I is regular, proving $\Omega = \Omega_X^p(\log D)$. $\square$

Each D_i may also denote a smooth divisor with several disjoint connected components, since locally it is still cut out by one parameter. Thus components carrying the same coefficient can be grouped together when their union is smooth.

Using the notation above, the residue map for one-forms is

$$\mathrm{Res}_D : \Omega_X^1(\log D) \longrightarrow \bigoplus_{i=1}^r \mathcal{O}_{D_i}, \quad \sum_{i=1}^n a_i \delta_i \longmapsto (a_i|_{D_i})_{1 \leq i \leq s}$$

near a point on exactly $D_1, \dots, D_s$. The other summands have zero stalk there. In this display Res_D denotes the usual total residue map. For $p \geq 1$, there is also

$$\mathrm{res}_{D_1} : \Omega_X^p(\log D) \longrightarrow \Omega_{D_1}^{p-1}(\log((D - D_1)|_{D_1})).$$

Locally write, with the logarithmic factor placed on the left,

$$\varphi = \varphi_1 + \frac{df_1}{f_1} \wedge \varphi_2, \quad \varphi_2 = \sum_{1 \in I} a_I \delta_{I \setminus \{1\}},$$

where φ_1 is spanned by the δ_I with $1 \notin I$. Then

$$\mathrm{res}_{D_1}(\varphi) = \varphi_2|_{D_1}.$$

The order of the wedge factors fixes the sign. The maps res_{D_i} are defined in the same way.

Finally, ordinary restriction gives

$$\rho_{D_1} : \Omega_X^p(\log(D - D_1)) \longrightarrow \Omega_{D_1}^p(\log((D - D_1)|_{D_1})).$$

The sheaf on the left has generators $f_1 \delta_I$ when $1 \in I$, and δ_I when $1 \notin I$. Thus

$$\rho_{D_1} \left(\sum_{1 \in I} f_1 a_I \delta_I + \sum_{1 \notin I} a_I \delta_I \right) = \sum_{1 \notin I} (a_I|_{D_1}) (\delta_I|_{D_1}).$$

Sheaves on D_i are regarded as sheaves on X by pushforward under the closed immersion.

Proposition 4.2.8. *We have the following exact sequences:*

$$(a) \quad 0 \longrightarrow \Omega_X^1 \longrightarrow \Omega_X^1(\log D) \xrightarrow{\mathrm{Res}_D} \bigoplus_{i=1}^r \mathcal{O}_{D_i} \longrightarrow 0.$$

(b) For $p \geq 1$,

$$\begin{aligned} 0 \longrightarrow \Omega_X^p(\log(D - D_1)) &\longrightarrow \Omega_X^p(\log D) \\ &\xrightarrow{\mathrm{res}_{D_1}} \Omega_{D_1}^{p-1}(\log((D - D_1)|_{D_1})) \longrightarrow 0. \end{aligned}$$

(c) For $p \geq 0$,

$$\begin{aligned} 0 \longrightarrow \Omega_X^p(\log D)(-D_1) &\longrightarrow \Omega_X^p(\log(D - D_1)) \\ &\xrightarrow{\rho_{D_1}} \Omega_{D_1}^p(\log((D - D_1)|_{D_1})) \longrightarrow 0. \end{aligned}$$

Proof. All assertions can be checked on the displayed local bases. In (a), vanishing residue means that the coefficient of df_i/f_i is divisible by f_i , so the form is regular. In (b), the coefficients of terms containing df_1/f_1 must be divisible by f_1 , which removes the pole along D_1 . Both residue maps are locally surjective.

For (c), restriction kills the terms $f_1\delta_I$ with $1 \in I$ because they contain df_1 . For $1 \notin I$, restriction vanishes precisely when the coefficient is divisible by f_1 . Thus the kernel has basis $f_1\delta_I$ for all I , which is exactly $\Omega_X^p(\log D)(-D_1)$. Restriction is locally surjective as well. $\square$

Corollary 4.2.9. *Assume D is a smooth divisor.*

(a) *We have exact sequences*

$$\begin{aligned} 0 \longrightarrow \Omega_X^p &\longrightarrow \Omega_X^p(\log D) \xrightarrow{\text{res}_D} \Omega_D^{p-1} \longrightarrow 0, \\ 0 \longrightarrow \Omega_X^p(\log D)(-D) &\longrightarrow \Omega_X^p \xrightarrow{\rho_D} \Omega_D^p \longrightarrow 0. \end{aligned}$$

In the first sequence we set $\Omega_D^{-1} = 0$ if $p = 0$.

(b) *Let $\pi : Y \rightarrow X$ be the cyclic covering of Proposition 4.1.8, and $D' \subseteq Y$ the divisor mapping isomorphically to D , with $\pi^*D = mD'$. Then pullback gives*

$$\pi^*\Omega_X^p(\log D) \cong \Omega_Y^p(\log D').$$

Proof. (a) follows from Proposition 4.2.8. For (b), locally $t^m = f$, so

$$\pi^*(df/f) = \frac{d(t^m)}{t^m} = m \frac{dt}{t}.$$

Since m is invertible, the logarithmic bases correspond. Away from the branch divisor the map is *étale*, giving the assertion there too. $\square$

4.2C Nakano Vanishing Theorem

Theorem 4.2.10 (Nakano vanishing). *Let X be a smooth complex projective variety of dimension n , and let A be ample. Then*

$$H^q(X, \Omega_X^p \otimes \mathcal{O}_X(A)) = 0 \quad (p + q > n).$$

Equivalently, by Serre duality,

$$H^s(X, \Omega_X^r \otimes \mathcal{O}_X(-A)) = 0 \quad (r + s < n),$$

where $s = n - q$ and $r = n - p$.

For $p = n$ this is Kodaira vanishing. The strategy is again to use a cyclic covering to represent the pulled-back ample divisor by a smooth effective divisor.

Proof. We prove the second statement by induction on n . For $n = 0$ there is nothing to prove; for $n = 1$, only $(r, s) = (0, 0)$ occurs, and $H^0(X, \mathcal{O}_X(-A)) = 0$ since $\deg(-A) < 0$.

Choose a smooth divisor $D \in |mA|$ by Bertini. We first claim that it suffices to prove

$$H^s(X, \Omega_X^r(\log D) \otimes \mathcal{O}_X(-A)) = 0 \quad (r + s < n). \quad (4.1)$$

Indeed, the first sequence in Corollary 4.2.9, twisted by $-A$, gives

$$0 \longrightarrow \Omega_X^r(-A) \longrightarrow \Omega_X^r(\log D)(-A) \longrightarrow \Omega_D^{r-1}(-A|_D) \longrightarrow 0.$$

By induction on D , the term $H^{s-1}(D, \Omega_D^{r-1}(-A|_D))$ vanishes whenever $r + s < n$. Hence the map from $H^s(X, \Omega_X^r(-A))$ to the group in (4.1) is injective. If $r = 0$ the sheaf with exterior degree -1 is zero; if $s = 0$ the negative cohomology group is zero.

For (4.1), take the m -fold cyclic covering $\pi : Y \rightarrow X$ branched along D . Lemma 4.1.18 reduces us to

$$H^s(Y, \pi^*(\Omega_X^r(\log D) \otimes \mathcal{O}_X(-A))) = 0.$$

Corollary 4.2.9(b) identifies $\pi^*\Omega_X^r(\log D)$ with $\Omega_Y^r(\log D')$, and the tautological section gives $D' \sim_{\text{lin}} \pi^*A$. Thus the group is

$$H^s(Y, \Omega_Y^r(\log D')(-D')).$$

The divisor D' is smooth and ample. By the second sequence in Corollary 4.2.9,

$$0 \longrightarrow \Omega_Y^r(\log D')(-D') \longrightarrow \Omega_Y^r \longrightarrow \Omega_{D'}^r \longrightarrow 0.$$

Lemma 4.2.4 makes the restriction in degree $s - 1$ surjective and the restriction in degree s injective, since $r + s < n$. Therefore the desired group vanishes. $\square$

4.3 Vanishing for Big and Nef Line Bundles

This section is devoted to the basic vanishing theorem, due to Kawamata and Viehweg, for big and nef divisors.

4.3A Statement and Proof of the Theorem

We can weaken the ampleness hypothesis in Kodaira vanishing to nefness and bigness. This result was independently discovered by Kawamata [Kaw82] and Viehweg [Vie82].

Theorem 4.3.1 (Kodaira vanishing for big and nef divisors). *Let X be a smooth complex projective variety of dimension n , and let D be a nef and big integral divisor on X . Then*

$$H^i(X, \mathcal{O}_X(K_X + D)) = 0 \quad (i > 0).$$

Equivalently, by Serre duality,

$$H^j(X, \mathcal{O}_X(-D)) = 0 \quad (j < n).$$

Remark 4.3.2. Recall from Theorem 3.2.15 that a nef divisor D is big if and only if its top self-intersection is strictly positive, i.e. $(D^n) > 0$. However, we do not use this characterization in the proof.

Remark 4.3.3. The papers [Kaw82, Theorem 2] and [Vie82, Proposition 2.3] also contain the fundamental vanishing theorem for $\mathbb{Q}$ -divisors, which we discuss in the next section.

Example 4.3.4. The analogue of Nakano vanishing can fail for big and nef divisors. Let $X = \text{Bl}_p \mathbb{P}^3$, and let $H \cong \mathbb{P}^2$ be the pullback of a general hyperplane not passing through p . Then H is nef and big, but

$$H^1(X, \Omega_X^1 \otimes \mathcal{O}_X(-H)) \neq 0.$$

Indeed, the blow-up formula [Voi02, 7.31] gives an isomorphism of complex vector spaces

$$H^k(X, \mathbb{C}) \cong H^k(\mathbb{P}^3, \mathbb{C}) \oplus H^{k-2}(\{p\}, \mathbb{C}) \oplus H^{k-4}(\{p\}, \mathbb{C}).$$

For $k = 2$ this gives $h^2(X) = 2$. Hodge decomposition also gives

$$H^2(X, \mathbb{C}) = H^{0,2}(X) \oplus H^1(X, \Omega_X^1) \oplus H^{2,0}(X).$$

The Hodge numbers $h^{p,0}$ are birational invariants of smooth projective varieties. Since projective space has only (p, p) -type cohomology, $h^{2,0}(X) = h^{0,2}(X) = 0$. Consequently,

$$\dim H^1(X, \Omega_X^1) = 2, \quad \dim H^1(H, \Omega_H^1) = 1.$$

The second sequence of Corollary 4.2.9 gives

$$\cdots \longrightarrow H^1(X, \Omega_X^1(\log H)(-H)) \longrightarrow H^1(X, \Omega_X^1) \xrightarrow{\varphi} H^1(H, \Omega_H^1) \longrightarrow \cdots.$$

Since $\dim \ker \varphi \geq 2 - 1 = 1$, the left-hand group is nonzero. The first sequence, twisted by $-H$, gives

$$\begin{aligned} \cdots \longrightarrow H^1(X, \Omega_X^1(-H)) &\longrightarrow H^1(X, \Omega_X^1(\log H)(-H)) \\ &\longrightarrow H^1(H, \mathcal{O}_H(-H)) \longrightarrow \cdots. \end{aligned}$$

The last group is $H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(-1)) = 0$. In fact, $H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(-1)) = 0$ too, so the first map in this display is an isomorphism. This proves the claimed nonvanishing.

The point is that, although the $h^{p,0}$ are birational invariants, the Hodge numbers $h^{p,q}$ for $p, q \geq 1$ need not be. Thus we cannot expect Nakano-type statements under these weaker positivity hypotheses. The term “plurigenus” instead refers to $h^0(X, mK_X)$.

We turn to the proof of Theorem 4.3.1. Following Kawamata’s original article, we reduce it to a statement that weakens positivity in the presence of a strong geometric hypothesis. This is a logarithmic version of Kodaira vanishing.

Lemma 4.3.5 (Norimatsu’s lemma). *Let X be a smooth projective variety, A an ample divisor, and E an snc divisor. Then*

$$H^i(X, \mathcal{O}_X(K_X + A + E)) = 0 \quad (i > 0).$$

Equivalently,

$$H^j(X, \mathcal{O}_X(-A - E)) = 0 \quad (j < \dim X).$$

Proof. Write $E = \sum_{i=1}^r E_i$. We induct on $n = \dim X$ and, for fixed n , on r . Dimension zero is immediate, and $r = 0$ is Kodaira vanishing. Consider

$$\begin{aligned} 0 \longrightarrow \mathcal{O}_X \left(-A - \sum_{i=1}^r E_i \right) &\longrightarrow \mathcal{O}_X \left(-A - \sum_{i=1}^{r-1} E_i \right) \\ &\longrightarrow \mathcal{O}_{E_r} \left(-A|_{E_r} - \sum_{i=1}^{r-1} E_i|_{E_r} \right) \longrightarrow 0. \end{aligned}$$

The restriction $A|_{E_r}$ is ample, and the sum of the nonempty $E_i|_{E_r}$ is snc on E_r . The induction on dimension gives vanishing of its cohomology in degrees $j - 1 < n - 1$. The induction on r gives vanishing of the middle term in degrees $j < n$. The long exact sequence therefore gives the desired vanishing of the left term for $j < n$. Negative cohomological degrees are understood to be zero. $\square$

Proof of Theorem 4.3.1. Since D is big, Kodaira’s lemma gives, for a suitable $m \gg 0$,

$$mD \sim_{\text{lin}} H + N, \tag{4.2}$$

where H is ample and N is effective. We proceed in two steps.

Step 1: We first reduce to the case where N has snc support. Take a log resolution $\mu : X^* \rightarrow X$ of N by smooth blow-ups. The divisor $D^* = \mu^* D$ remains nef and big. Corollary 4.1.6 gives

$$\mu_* \mathcal{O}_{X^*}(K_{X^*}) = \mathcal{O}_X(K_X), \quad R^i \mu_* \mathcal{O}_{X^*}(K_{X^*}) = 0 \quad (i > 0). \tag{4.3}$$

The pullback μ^*H is not necessarily ample, so we modify the decomposition. Write

$$\mu^*N = \sum_j a_j F_j, \quad a_j \geq 0,$$

including all exceptional divisors among the F_j , with zero coefficients where necessary. The reduced union of these divisors is snc. By Corollary 4.1.6(ii), for $p \gg 0$ there are $b_j \geq 0$, positive on the exceptional divisors and zero on the others, such that

$$\mu^*(pH) - \sum_j b_j F_j$$

is ample. Hence

$$\mu^*(pmD) \sim_{\text{lin}} \left(\mu^*(pH) - \sum_j b_j F_j \right) + \sum_j (pa_j + b_j) F_j.$$

This is the promised decomposition $m'D^* \sim_{\text{lin}} H^* + N^*$, with H^* ample and N^* effective with snc support.

Suppose the vanishing is known on X^* . Both assertions in (4.3), the projection formula, and the Leray spectral sequence give

$$\begin{aligned} H^i(X^*, \mathcal{O}_{X^*}(K_{X^*} + \mu^*D)) &\cong H^i(X, \mu_* \mathcal{O}_{X^*}(K_{X^*} + \mu^*D)) \\ &\cong H^i(X, \mathcal{O}_X(K_X + D)). \end{aligned}$$

Thus vanishing descends to X . We may therefore assume that N in (4.2) has snc support.

Step 2: Write $N = \sum_{i=1}^r e_i E_i$, with $e_i > 0$ integral. If $N = 0$, then $mD \sim_{\text{lin}} H$ makes D ample, and Kodaira vanishing applies. Otherwise put

$$e^* = e_1 \cdots e_r, \quad e_i^* = e^* / e_i.$$

Use the Kawamata covering with integers $me_1^*, \dots, me_r^*$ to obtain $h : Y \rightarrow X$, with Y smooth, such that

$$h^*E_i = me_i^*E'_i, \quad E' = \sum_i E'_i \text{ is snc.}$$

Let $D' = h^*D$ and $H' = h^*H$. Then

$$mD' \sim_{\text{lin}} H' + me^*E'. \tag{4.4}$$

The divisor D' is nef, and H' is ample since h is finite. Consequently,

$$me^*(D' - E') \sim_{\text{lin}} H' + m(e^* - 1)D'$$

is ample, being a sum of an ample and a nef divisor. Thus $A' = D' - E'$ is ample, and Norimatsu's lemma gives

$$H^j(Y, \mathcal{O}_Y(-D')) = H^j(Y, \mathcal{O}_Y(-A' - E')) = 0 \quad (j < n).$$

Finally, Lemma 4.1.18 gives

$$H^j(X, \mathcal{O}_X(-D)) \hookrightarrow H^j(Y, \mathcal{O}_Y(-D')) = 0,$$

as required. $\square$

Example 4.3.6 (An extension of Theorem 4.3.1). Let D be a nef divisor on a smooth projective n -fold X . Suppose

$$(D^{n-k} \cdot H^k) > 0$$

for some integer $0 \leq k \leq n$ and an ample divisor H . Then

$$H^i(X, \mathcal{O}_X(K_X + D)) = 0 \quad (i > k).$$

This statement is also due to Kawamata and Viehweg and contains Theorem 4.3.1 as the case $k = 0$.

Proof. For $k = 0$, the nef divisor D is big by its positive top intersection, so Theorem 4.3.1 applies. For $k = n$, the result is Grothendieck vanishing. It remains to consider $1 \leq k \leq n - 1$.

After replacing H by a sufficiently high multiple, choose a smooth very ample representative, still denoted H . Induct on n . On H , the nef divisor $D|_H$ satisfies the condition with $k' = k - 1$, since

$$((D|_H)^{(n-1)-(k-1)} \cdot (H|_H)^{k-1})_H = (D^{n-k} \cdot H^k)_X > 0.$$

Adjunction gives

$$0 \longrightarrow \mathcal{O}_X(K_X + D) \longrightarrow \mathcal{O}_X(K_X + D + H) \longrightarrow \mathcal{O}_H(K_H + D|_H) \longrightarrow 0.$$

The middle term has no higher cohomology by Kodaira vanishing, since $D + H$ is ample. For $i > k$, the induction hypothesis gives

$$H^{i-1}(H, \mathcal{O}_H(K_H + D|_H)) = 0$$

because $i - 1 > k - 1$. The long exact sequence proves the assertion. The dimension-one cases were already covered by $k = 0$ and $k = n$. $\square$

We give two theorems without proof.

Theorem 4.3.7 (Kollár's injectivity theorem). *Let X be smooth and projective, and L a semi-ample divisor. Given $k \geq 1$, fix $D \in |kL|$. The homomorphisms given by multiplication by its defining section,*

$$H^i(X, \mathcal{O}_X(K_X + mL)) \xrightarrow{\cdot D} H^i(X, \mathcal{O}_X(K_X + (m+k)L)),$$

are injective for all $i \geq 0$ and $m > 0$.

This implies Kodaira vanishing: if L is ample, take $m = 1$ and $k \gg 0$ with $|kL| \neq \emptyset$. The target is $H^i(X, \mathcal{O}_X(K_X + (k+1)L))$, which vanishes for $i > 0$ by Serre vanishing.

The second theorem shows that the direct images of the canonical bundle have exceptionally good cohomological properties.

Theorem 4.3.8 (Kollár vanishing theorem). *Let $f : X \rightarrow Y$ be a surjective morphism of projective varieties, with X smooth. Then:*

- (i) $R^j f_* \mathcal{O}_X(K_X)$ is torsion-free for every $j \geq 0$.
- (ii) $R^j f_* \mathcal{O}_X(K_X) = 0$ for $j > \dim X - \dim Y$.
- (iii) For every ample divisor A on Y and every $j \geq 0$,

$$H^i(Y, R^j f_* \mathcal{O}_X(K_X) \otimes \mathcal{O}_Y(A)) = 0 \quad (i > 0).$$

We used this result in the proof of Fujita's vanishing theorem.

4.3B Some Applications

Theorem 4.3.9 (Grauert–Riemenschneider vanishing). *Let $f : X \rightarrow Y$ be a generically finite, surjective projective morphism of varieties, with X smooth. Then*

$$R^i f_* \mathcal{O}_X(K_X) = 0 \quad (i > 0).$$

Proof. See [Laz04a, 4.3.9] for a proof. We will also obtain this as a special case of generalized Kawamata–Viehweg vanishing at the end of §4.4. $\square$

4.4 Kawamata–Viehweg Vanishing Theorem

We now come to the basic vanishing theorem for $\mathbb{Q}$ -divisors. First we need some operations on divisors.

Definition 4.4.1 (Rounds). Let $D = \sum_i a_i D_i$ be a $\mathbb{Q}$ -divisor, with distinct prime divisors D_i . Its round-up and round-down are the integral divisors

$$\lceil D \rceil = \sum_i \lceil a_i \rceil D_i, \quad \lfloor D \rfloor = \sum_i \lfloor a_i \rfloor D_i,$$

where $\lceil x \rceil$ is the least integer at least x , and $\lfloor x \rfloor$ the greatest integer at most x . The fractional part is

$$\{D\} = D - \lfloor D \rfloor.$$

The same definitions apply coefficientwise to $\mathbb{R}$ -divisors.

Remark 4.4.2. Rounding is defined for actual divisors, at the cycle level, rather than for their linear or numerical equivalence classes. On a smooth variety, Cartier and Weil divisors are identified, so rounding a $\mathbb{Q}$ -Cartier divisor in this sense does make sense and gives an integral Cartier divisor. On a singular variety, however, its round-up or round-down need not be Cartier; additional hypotheses may be needed for pullbacks and other operations.

Warning 4.4.3 (Warning on rounds). Pullback does not commute with rounding, and numerical equivalence does not determine the rounding. For example, in $X = \mathbb{A}^2$ let $A = V(y - x^2)$, let Y be the x -axis, and let P be the origin. Then Y is tangent to A at P . For $D = \frac{1}{2}A$,

$$\lfloor D \rfloor|_Y = 0, \quad \lfloor D|_Y \rfloor = \lfloor \tfrac{1}{2} \cdot 2P \rfloor = P.$$

Likewise, if $L, C \subseteq \mathbb{P}^2$ are a line and an irreducible conic, then $L \equiv_{\text{num}} \frac{1}{2}C$, but

$$\lfloor \tfrac{1}{2}C \rfloor = 0, \quad \lfloor L \rfloor = L.$$

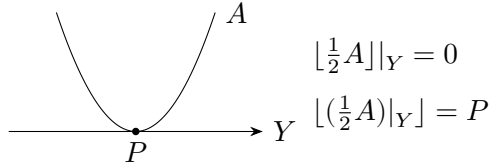


Figure 4.4: Rounding does not commute with restriction.

4.4A Statement and Proof of the Theorem

Theorem 4.4.4 (Kawamata–Viehweg vanishing). *Let X be a non-singular projective variety of dimension n , and N an integral divisor on X . Assume*

$$N \equiv_{\text{num}} B + \Delta,$$

where B is a nef and big $\mathbb{Q}$ -divisor and $\Delta = \sum_i a_i \Delta_i$ has snc support, with $0 \leq a_i < 1$. Then

$$H^i(X, \mathcal{O}_X(K_X + N)) = 0 \quad (i > 0).$$

Equivalently, by Serre duality,

$$H^j(X, \mathcal{O}_X(-N)) = 0 \quad (j < n).$$

Remark 4.4.5. The hypothesis on Δ says that it is effective, its support is snc, and $\lfloor \Delta \rfloor = 0$. In standard terminology a boundary has coefficients in $[0, 1]$; here we require all coefficients to be strictly less than 1. This strict inequality is part of the vanishing hypothesis.

Proof of Theorem 4.4.4. We allow each Δ_i to have several disjoint smooth connected components, as in the discussion following Proposition 4.2.7. This is useful because the pullback of a point on a curve may be disconnected. We induct on the number of nonzero fractional terms.

If $\Delta = 0$, then N is nef and big because these properties are numerical, and Theorem 4.3.1 applies. Otherwise write $a_1 = c/d$, with integers $0 < c < d$. By Lemma 4.1.18, it suffices to find a finite surjective morphism $f : X' \rightarrow X$, with X' non-singular, for which

$$H^j(X', f^* \mathcal{O}_X(-N)) = 0 \quad (j < n). \quad (4.5)$$

Take a BG covering such that $\Delta'_i = f^* \Delta_i$ are smooth, their sum is snc, and

$$\Delta'_1 \sim_{\text{lin}} dA' \quad (4.6)$$

for some divisor A' on X' . Set $N' = f^*N$ and $B' = f^*B$.

Next take the d -fold cyclic covering $\pi : X'' \rightarrow X'$ branched along Δ'_1 . Its root divisor Δ''_1 satisfies

$$d\Delta''_1 = \pi^* \Delta'_1, \quad \Delta''_1 \sim_{\text{lin}} A'', \quad A'' = \pi^* A'.$$

The linear equivalence comes from the tautological section. For $i \geq 2$, put $\Delta''_i = \pi^* \Delta'_i$, and set $N'' = \pi^* N'$, $B'' = \pi^* B'$. Then

$$N'' \equiv_{\text{num}} B'' + cA'' + \sum_{i \geq 2} a_i \Delta''_i,$$

so

$$N'' - cA'' \equiv_{\text{num}} B'' + \sum_{i \geq 2} a_i \Delta''_i.$$

The divisor B'' is nef and big, and the remaining fractional part has snc support by Proposition 4.1.12. The induction hypothesis gives

$$H^j(X'', \mathcal{O}_{X''}(-N'' + cA'')) = 0 \quad (j < n),$$

and, since π is finite,

$$H^j(X', \pi_* \mathcal{O}_{X''}(-N'' + cA'')) = 0 \quad (j < n). \quad (4.7)$$

By Remark 4.1.10 and the projection formula,

$$\begin{aligned} \pi_* \mathcal{O}_{X''}(-N'' + cA'') &= (\pi_* \mathcal{O}_{X''}) \otimes \mathcal{O}_{X'}(-N' + cA') \\ &\cong \bigoplus_{a=0}^{d-1} \mathcal{O}_{X'}(-N' + (c-a)A'). \end{aligned}$$

The summand with $a = c$ is $\mathcal{O}_{X'}(-N')$, since $1 \leq c \leq d-1$. Thus (4.7) implies (4.5), proving the theorem. $\square$

Corollary 4.4.6. *Let X be smooth and projective, and B a nef and big $\mathbb{Q}$ -divisor whose fractional part has snc support. Then*

$$H^i(X, \mathcal{O}_X(K_X + \lceil B \rceil)) = 0 \quad (i > 0).$$

Proof. Since $\lceil B \rceil = -\lfloor -B \rfloor$, we have

$$B - \lceil B \rceil = B + \lfloor -B \rfloor = -\{-B\}.$$

Thus $\lceil B \rceil = B + \{-B\}$, where $\{-B\}$ has the same support as $\{B\}$ and all coefficients less than 1. Apply Theorem 4.4.4. $\square$

Corollary 4.4.7. *Let X be smooth and projective, L an integral divisor, and D a $\mathbb{Q}$ -divisor. If $L - D$ is nef and big and D has snc support, then*

$$H^i(X, \mathcal{O}_X(K_X + L - \lfloor D \rfloor)) = 0 \quad (i > 0).$$

Proof. Use $L - \lfloor D \rfloor = (L - D) + \{D\}$ and Theorem 4.4.4. $\square$

4.4B Generalizations of the Vanishings in Relative Version

Following [Kaw24] and [KMM87], we generalize the vanishing theorem in directions needed for higher-dimensional algebraic geometry. Recall that an $\mathbb{R}$ -Cartier divisor is f -nef if its restriction to every fiber is nef, and f -big if its restriction to the generic fiber is big. We first need a lemma.

Lemma 4.4.8. *Let X be smooth, and let D be an $\mathbb{R}$ -divisor such that $\{D\}$ has snc support. If $f : Y \rightarrow X$ is a proper birational morphism with Y smooth, then*

$$f_* \mathcal{O}_Y(K_Y + \lceil f^* D \rceil) \cong \mathcal{O}_X(K_X + \lceil D \rceil).$$

Proof. Choose compatible canonical divisors. Put $\Delta = \lceil D \rceil - \lfloor D \rfloor$, the reduced snc divisor supported on the nonintegral components of D . The discrepancy formula gives

$$K_Y + f_*^{-1} \Delta = f^*(K_X + \Delta) + \sum_{E_i \text{ exceptional}} a_i E_i.$$

For a reduced snc pair the coefficients a_i are integers at least -1 ; see the local snc calculation in Lemma 5.2.20. We only need this lower bound, not equality of the infimum with -1 .

Adding $\lceil f^* D \rceil$ and rearranging, using that $\lfloor D \rfloor$ is integral Cartier, gives

$$K_Y + \lceil f^* D \rceil = f^*(K_X + \lceil D \rceil) + F,$$

where

$$F = \lceil f^* \{D\} \rceil - f_*^{-1} \Delta + \sum_i a_i E_i.$$

The coefficients on nonexceptional divisors cancel, since $[\{D\}] = \Delta$. For an exceptional divisor E_i , its coefficient is

$$\text{mult}_{E_i}[\{f^*\{D\}\}] + a_i.$$

If $a_i \geq 0$, this is nonnegative because $\{D\}$ is effective. If $a_i = -1$, its center lies in $\text{Supp } \Delta$: outside that divisor, the relative canonical divisor of a birational morphism between smooth varieties is effective. Since every component of Δ has a strictly positive coefficient in $\{D\}$, we then have

$$\text{mult}_{E_i}(f^*\{D\}) > 0, \quad \text{mult}_{E_i}[\{f^*\{D\}\}] \geq 1.$$

Thus this coefficient is again nonnegative. Hence F is effective and exceptional. Algebraic Hartogs gives $f_*\mathcal{O}_Y(F) = \mathcal{O}_X$, and the projection formula yields

$$\begin{aligned} f_*\mathcal{O}_Y(K_Y + [f^*D]) &\cong \mathcal{O}_X(K_X + [D]) \otimes f_*\mathcal{O}_Y(F) \\ &\cong \mathcal{O}_X(K_X + [D]). \end{aligned}$$

□

Theorem 4.4.9 (Generalized Kodaira vanishing in the projective case). *Let X be a smooth projective variety, $f: X \rightarrow S$ a surjective morphism onto a projective variety, and D an f -ample $\mathbb{Q}$ -divisor with $\{D\}$ of snc support. Then*

$$R^i f_*\mathcal{O}_X(K_X + [D]) = 0 \quad (i > 0).$$

Proof. Step 1: We can add the pullback of any integral Cartier divisor A on S . Indeed,

$$\begin{aligned} R^i f_*\mathcal{O}_X(K_X + [D + f^*A]) &= R^i f_*\mathcal{O}_X(K_X + [D] + f^*A) \\ &\cong R^i f_*\mathcal{O}_X(K_X + [D]) \otimes \mathcal{O}_S(A). \end{aligned}$$

Vanishing is therefore unchanged.

Step 2: Clear denominators so that rD is an integral f -ample divisor on X . Choose an ample integral Cartier divisor A on S and replace it by a sufficiently high multiple. Relative ampleness then makes $D + f^*A$ ample on X . Merely increasing the common denominator does not do this; the multiple of A must be large enough.

Replace D by $D + f^*A$. Its fractional part is unchanged. Fix an ample divisor H on S and consider the Leray spectral sequence

$$\begin{aligned} E_2^{p,q} = H^p(S, R^q f_*\mathcal{O}_X(K_X + [D]) \otimes \mathcal{O}_S(mH)) &\implies \\ H^{p+q}(X, \mathcal{O}_X(K_X + [D] + mf^*H)). \end{aligned}$$

For $m \gg 0$, Serre vanishing makes $E_2^{p,q} = 0$ for $p > 0$. There are only finitely many nonzero higher direct images, so the same m works for all q . Thus

$$H^0(S, R^i f_* \mathcal{O}_X(K_X + \lceil D \rceil) \otimes \mathcal{O}_S(mH)) \cong H^i(X, \mathcal{O}_X(K_X + \lceil D \rceil + mf^*H)).$$

Since D is ample and f^*H is nef, $D + mf^*H$ is ample. Corollary 4.4.6 shows that the right side vanishes for $i > 0$. For $m \gg 0$, Serre's global generation theorem also makes

$$R^i f_* \mathcal{O}_X(K_X + \lceil D \rceil) \otimes \mathcal{O}_S(mH)$$

globally generated. A globally generated sheaf with no global sections is zero. Twisting back proves the assertion. $\square$

Theorem 4.4.10 (Generalized Kodaira vanishing in the general case). *Let X be smooth, $f : X \rightarrow S$ a projective surjective morphism onto a variety, and D an f -ample $\mathbb{Q}$ -divisor whose fractional part has snc support. Then*

$$R^i f_* \mathcal{O}_X(K_X + \lceil D \rceil) = 0 \quad (i > 0).$$

The statement also applies when f is proper and admits the given relatively ample divisor: such a morphism is projective, as in Remark 1.5.2.

Proof. The idea is to turn the problem into the projective case. The assertion is local on S , so assume S affine. Take a projective compactification $\bar{S}$ and, using a projective embedding of X over S , take the closure of X over $\bar{S}$. Resolve outside X to obtain a smooth projective $\bar{X}$ and a projective morphism $\bar{f} : \bar{X} \rightarrow \bar{S}$ fitting into

$$\begin{array}{ccc} X & \hookrightarrow & \bar{X} \\ f \downarrow & & \downarrow \bar{f} \\ S & \hookrightarrow & \bar{S}. \end{array}$$

The square is Cartesian: $X = \bar{f}^{-1}(S)$. This property is preserved by the modifications outside X below.

Step 1: We construct a projective birational morphism $\bar{\mu} : \bar{Y} \rightarrow \bar{X}$, with $\bar{Y}$ smooth, and a $\mathbb{Q}$ -divisor $\bar{C}$ such that:

- (1) $\bar{C}$ is nef and big over $\bar{S}$;
- (2) $\text{Supp}\{\bar{C}\} \cup \text{Exc}(\bar{\mu})$ is contained in an snc divisor;
- (3) on $Y = \bar{\mu}^{-1}(X)$, we have $\bar{C}|_Y = \mu^*D$, where $\mu = \bar{\mu}|_Y$.

Lemma 4.4.8 will then give

$$\mu_* \mathcal{O}_Y(K_Y + \lceil \mu^*D \rceil) \cong \mathcal{O}_X(K_X + \lceil D \rceil). \quad (4.8)$$

Let $\bar{D}$ be the closure of D in $\bar{X}$. Choose m sufficiently divisible and large so that $m\bar{D}$ is integral and mD is f -very ample. Consider evaluation

$$\varphi_m : \bar{f}^* \bar{f}_* \mathcal{O}_{\bar{X}}(m\bar{D}) \longrightarrow \mathcal{O}_{\bar{X}}(m\bar{D}).$$

It is surjective on X . Its image has the form $\mathcal{I} \mathcal{O}_{\bar{X}}(m\bar{D})$, with $\mathcal{I}|_X = \mathcal{O}_X$. Blow up this ideal and resolve, only outside X . Replacing the compactification and pulling back $\bar{D}$, we obtain an effective Cartier divisor E supported outside X , and an invertible quotient

$$\bar{f}^* \mathcal{E} \twoheadrightarrow \mathcal{O}_{\bar{X}}(m\bar{D} - E),$$

where $\mathcal{E}$ is the coherent sheaf on $\bar{S}$ used in the original evaluation. This gives an $\bar{S}$ -morphism

$$\pi : \bar{X} \longrightarrow \mathbb{P}_{\bar{S}}(\mathcal{E}).$$

Let $\bar{P}$ be its reduced image. The relatively ample line bundle $\mathcal{O}_{\bar{P}}(1)$ corresponds to a divisor H , and

$$\pi^* H \sim_{\text{lin}} m\bar{D} - E.$$

On X , this is the map given by the complete relative linear system of mD , so $\pi|_X$ is a closed immersion, in particular finite. One can also see finiteness from positivity: a contracted complete curve $C \subseteq X$ would satisfy

$$(mD \cdot C) = (H \cdot \pi_* C) = 0,$$

contrary to relative ampleness, since π is over S . A positive-dimensional projective fiber would contain such a curve. Thus π is generically finite, and

$$D_0 = \bar{D} - \frac{1}{m} E \sim_{\mathbb{Q}} \frac{1}{m} \pi^* H$$

is $\bar{f}$ -nef and $\bar{f}$ -big, with $D_0|_X = D$.

Now resolve the support of $\{D_0\}$ together with the exceptional locus. Set $\bar{C} = \bar{\mu}^* D_0$. This gives (1)–(3), since the pullback of the integral part of D_0 is integral, and any new fractional components are among the exceptional divisors.

Step 2: The divisor $\bar{C}$ is only relatively nef and big; Theorem 4.4.9 cannot yet be applied to it. We use the small-coefficient perturbation construction of Lemma 4.4.13 below, over the projective base $\bar{S}$. That construction uses only Chow's lemma, Kodaira's lemma, resolution, and relative ampleness, so this forward reference does not use the vanishing theorem being proved.

It gives a projective birational morphism $\nu : \bar{Z} \rightarrow \bar{Y}$, with $\bar{Z}$ smooth, and an effective divisor $\sum \delta_j F_j$ with arbitrarily small positive rational coefficients such that

$$\bar{A} = \nu^* \bar{C} - \sum_j \delta_j F_j$$

is ample over $\bar{S}$ and has fractional part of snc support. Choose the coefficients small enough that

$$\lceil \bar{A} \rceil = \lceil \nu^* \bar{C} \rceil.$$

This is possible coefficient by coefficient: subtract less than the fractional part from a nonintegral coefficient, and less than 1 from an integral coefficient.

Put $g = \bar{\mu} \circ \nu : \bar{Z} \rightarrow \bar{X}$ and $h = \bar{f} \circ g : \bar{Z} \rightarrow \bar{S}$. Since $\bar{A}$ is h -ample, it is also g -ample. Both bases and $\bar{Z}$ are projective. Applying Theorem 4.4.9 to g and h gives, for

$$\begin{aligned} \mathcal{G} &= \mathcal{O}_{\bar{Z}}(K_{\bar{Z}} + \lceil \bar{A} \rceil), \\ R^q g_* \mathcal{G} &= 0 \quad (q > 0), \quad R^i h_* \mathcal{G} = 0 \quad (i > 0). \end{aligned}$$

The spectral sequence

$$R^p \bar{f}_*(R^q g_* \mathcal{G}) \implies R^{p+q} h_* \mathcal{G}$$

therefore collapses to the p -axis, giving

$$R^i \bar{f}_*(g_* \mathcal{G}) = 0 \quad (i > 0).$$

On $Z = g^{-1}(X)$, write $g_X = g|_Z$. We have $\nu^* \bar{C}|_Z = g_X^* D$, so Lemma 4.4.8 gives

$$(g_* \mathcal{G})|_X = (g_X)_* \mathcal{O}_Z(K_Z + \lceil g_X^* D \rceil) \cong \mathcal{O}_X(K_X + \lceil D \rceil).$$

Restriction to the open subset $S \subseteq \bar{S}$ commutes with higher direct images. Thus

$$0 = (R^i \bar{f}_*(g_* \mathcal{G}))|_S \cong R^i f_* \mathcal{O}_X(K_X + \lceil D \rceil),$$

which is the desired vanishing. $\square$

The relatively ample statement extends to $\mathbb{R}$ -divisors by perturbing coefficients.

Theorem 4.4.11 (Generalized Kodaira vanishing for $\mathbb{R}$ -divisors). *Let X be smooth, $f : X \rightarrow S$ projective and surjective, and D an f -ample $\mathbb{R}$ -divisor whose fractional part has snc support. Then*

$$R^i f_* \mathcal{O}_X(K_X + \lceil D \rceil) = 0 \quad (i > 0).$$

Proof. By openness of the relative ample cone, we can approximate the finitely many coefficients of D by rational numbers while preserving relative ampleness. Keep the integral coefficients fixed, and keep each nonintegral coefficient strictly between its original floor and ceiling. The resulting $\mathbb{Q}$ -divisor has the same round-up and the same fractional support. Theorem 4.4.10 applies. $\square$

The following is the relative Kawamata–Viehweg theorem; see [KMM87, Theorem 1-2-3].

Theorem 4.4.12 (Generalized Kawamata–Viehweg vanishing). *Let X be non-singular, $f : X \rightarrow S$ a proper surjective morphism, and D an f -nef and f -big $\mathbb{Q}$ -divisor whose fractional part has snc support. Then*

$$R^i f_* \mathcal{O}_X(K_X + \lceil D \rceil) = 0 \quad (i > 0).$$

Before proving this, we give the perturbation lemma used above. The general statement is [KMM87, Corollary 0-3-6]. Its construction is independent of the vanishing theorems.

Lemma 4.4.13. *Let $f : X \rightarrow S$ be a proper surjective morphism, with X normal, and let D be an f -nef and f -big $\mathbb{Q}$ -Cartier divisor. There exist a projective birational morphism $\mu : Y \rightarrow X$, with Y smooth and projective over S , and finitely many prime divisors F_j on Y , such that*

$$\text{Supp}(\mu^* D) \cup \bigcup_j F_j \cup \text{Exc}(\mu)$$

is contained in an snc divisor, and

$$\mu^* D - \sum_j \delta_j F_j$$

is $(f \circ \mu)$ -ample for suitable positive rational coefficients δ_j that can be made arbitrarily small on the chosen model.

Construction for the bases used here. We give the construction when S is quasi-projective. This covers both the projective base in the proof of Theorem 4.4.10 and every affine neighborhood needed to prove the relative vanishing statements. The form over an arbitrary algebraic base is the relative perturbation lemma cited above; the local construction alone is not a gluing argument for one global model over such a base.

Step 1: By Chow’s lemma, choose a variety X' projective over S and a proper birational morphism $\varphi : X' \rightarrow X$. It is projective: its graph is a closed immersion into $X' \times_S X$, which is projective over X , since X is separated over S . Normalize X' and compose with the finite normalization map. We retain the notation X' , φ , and $f' : X' \rightarrow S$.

Step 2: The divisor $D_0 = \varphi^* D$ is f' -nef and f' -big. Relative Kodaira’s lemma gives

$$mD_0 \sim_S A' + N',$$

where A' is f' -ample and integral, and N' is effective. For completeness, choose such an A' first. Kodaira’s lemma on the generic fiber makes $f'_* \mathcal{O}_{X'}(mD_0 - A')$ nonzero for a suitable divisible m . Since S is quasi-projective, twisting this coherent sheaf by a sufficiently ample line bundle from S gives a nonzero global section. Its divisor gives N' , with the twist from S absorbed by $\sim_S$.

Set $E = N'/m$. For any rational $0 < \varepsilon < 1$,

$$D_0 \sim_{S, \mathbb{Q}} (1 - \varepsilon)D_0 + \frac{\varepsilon}{m}A' + \varepsilon E.$$

The divisor

$$A_\varepsilon = (1 - \varepsilon)D_0 + \frac{\varepsilon}{m}A'$$

is f' -ample, being a sum of a relatively nef and a relatively ample divisor. Thus $D_0 \sim_{S, \mathbb{Q}} A_\varepsilon + \varepsilon E$, with the coefficients of εE arbitrarily small.

Step 3: Take a log resolution $\mu' : Y \rightarrow X'$ of the *union* of the supports of D_0 and E , including the exceptional loci in the construction. We can arrange that $\text{Supp } \mu'^*D_0$, $\text{Supp } \mu'^*E$ and $\text{Exc}(\varphi \circ \mu')$ are contained in an snc divisor. Resolving the union is necessary: components of the sum $D_0 + E$ could cancel.

$$\begin{array}{ccccc} Y & \xrightarrow{\mu'} & X' & \xrightarrow{\varphi} & X \\ & & \downarrow f' & & \downarrow f \\ & & S & = & S. \end{array}$$

The resolution can be made by blow-ups, and there is an effective exceptional Cartier divisor F with $-F$ μ' -ample. This follows by combining the relatively ample $\mathcal{O}(1) = \mathcal{O}(-E_i)$ of the blow-ups, as in the induction in Corollary 4.1.6(ii). For a sufficiently small rational $\xi > 0$,

$$\mu'^*A_\varepsilon - \xi F$$

is ample over S : a sufficiently large multiple of the pullback of an f' -ample divisor, minus F , is ample over S . Consequently,

$$\mu'^*D_0 - (\mu'^*(\varepsilon E) + \xi F) \sim_{S, \mathbb{Q}} \mu'^*A_\varepsilon - \xi F$$

is ample over S .

Step 4: Set $\mu = \varphi \circ \mu'$, so $\mu^*D = \mu'^*D_0$, and write

$$\sum_j \delta_j F_j = \mu'^*(\varepsilon E) + \xi F.$$

The support is contained in the snc divisor constructed in Step 3. Fix the resolution first, choose ε small, and then choose ξ sufficiently small for ampleness. This makes all the positive coefficients δ_j as small as required, and proves the asserted construction in the cases used here. $\square$

Proof of Theorem 4.4.12. Work locally on S , so assume S affine. Lemma 4.4.13 provides $\mu : Y \rightarrow X$, with Y smooth and projective over S , and an $(f \circ \mu)$ -ample divisor

$$P = \mu^*D - \sum_j \delta_j F_j$$

with snc fractional support. Choose the coefficients small enough that $\lceil P \rceil = \lceil \mu^* D \rceil$. There are only finitely many coefficients to check: at a nonintegral coefficient a , subtract less than $a - \lfloor a \rfloor$; at an integral coefficient, subtract less than 1.

The divisor P is also μ -ample. Applying Theorem 4.4.10 to μ gives

$$R^i \mu_* \mathcal{O}_Y(K_Y + \lceil P \rceil) = R^i \mu_* \mathcal{O}_Y(K_Y + \lceil \mu^* D \rceil) = 0 \quad (i > 0). \quad (4.9)$$

Lemma 4.4.8 identifies the degree-zero direct image with $\mathcal{O}_X(K_X + \lceil D \rceil)$. The spectral sequence

$$R^p f_* (R^q \mu_* \mathcal{O}_Y(K_Y + \lceil P \rceil)) \implies R^{p+q} (f \circ \mu)_* \mathcal{O}_Y(K_Y + \lceil P \rceil)$$

collapses to the p -axis by (4.9). Applying Theorem 4.4.10 to $f \circ \mu$, we conclude, for $i > 0$,

$$\begin{aligned} 0 &= R^i (f \circ \mu)_* \mathcal{O}_Y(K_Y + \lceil P \rceil) \\ &\cong R^i f_* (\mu_* \mathcal{O}_Y(K_Y + \lceil \mu^* D \rceil)) \\ &\cong R^i f_* \mathcal{O}_X(K_X + \lceil D \rceil). \end{aligned}$$

These local vanishings prove the assertion on S . □

Theorem 4.4.14 (Generalized Kawamata–Viehweg vanishing for $\mathbb{R}$ -divisors). *Let X be non-singular, $f : X \rightarrow S$ proper and surjective, and D an f -nef and f -big $\mathbb{R}$ -divisor whose fractional part has snc support. Then*

$$R^i f_* \mathcal{O}_X(K_X + \lceil D \rceil) = 0 \quad (i > 0).$$

Proof. Work over an affine open subset of S . The crucial point is the real-divisor version of the construction in Lemma 4.4.13. After the projective modification, relative Kodaira’s lemma for real divisors gives

$$D_0 \sim_{S, \mathbb{R}} A_0 + E,$$

where A_0 is f' -ample and E is effective. This form follows by expressing a big real divisor as a positive combination of nearby rational big divisors and applying the rational relative Kodaira lemma to each of them. We do not approximate a nef divisor and assume the approximations remain nef.

For a small real $0 < \varepsilon < 1$, put

$$A_\varepsilon = (1 - \varepsilon)D_0 + \varepsilon A_0.$$

This is f' -ample, and $D_0 \sim_{S, \mathbb{R}} A_\varepsilon + \varepsilon E$. Resolve the union of supports as above and choose $-F$ relatively ample for μ' . We need

$$\mu'^* D_0 - (\mu'^*(\varepsilon E) + \xi F) \sim_{S, \mathbb{R}} \mu'^* A_\varepsilon - \xi F$$

to be ample over S for small $\xi > 0$.

Write the ample real divisor as a *positive* linear combination

$$A_\varepsilon = \sum_i r_i H_i, \quad r_i > 0,$$

where the H_i are f' -ample integral divisors. For each i , the divisor $n\mu'^*H_i - F$ is ample over S for all sufficiently large integers n . Choose one such n for every i . Then

$$\sum_i r_i \left(\mu'^*H_i - \frac{1}{n}F \right) = \mu'^*A_\varepsilon - \frac{\sum_i r_i}{n}F$$

is ample over S . Taking $\xi = (\sum_i r_i)/n$ makes ξ arbitrarily small. Thus the desired effective perturbation $\sum_j \delta_j F_j = \mu'^*(\varepsilon E) + \xi F$ exists with arbitrarily small real coefficients.

Choose them so that rounding is unchanged. Apply Theorem 4.4.11 to μ and to $f \circ \mu$, and then use Lemma 4.4.8 and the same spectral sequence as for Theorem 4.4.12. This proves the real-divisor statement. $\square$

We can now finish the proof of Grauert–Riemenschneider vanishing.

Proof of Theorem 4.3.9. Since f is generically finite, its generic fiber has dimension zero. The divisor $D = 0$ is f -nef, and it is f -big because its restriction to that zero-dimensional generic fiber is big. Its fractional part is empty. Theorem 4.4.12 therefore gives

$$R^i f_* \mathcal{O}_X(K_X) = R^i f_* \mathcal{O}_X(K_X + \lceil 0 \rceil) = 0 \quad (i > 0).$$

$\square$

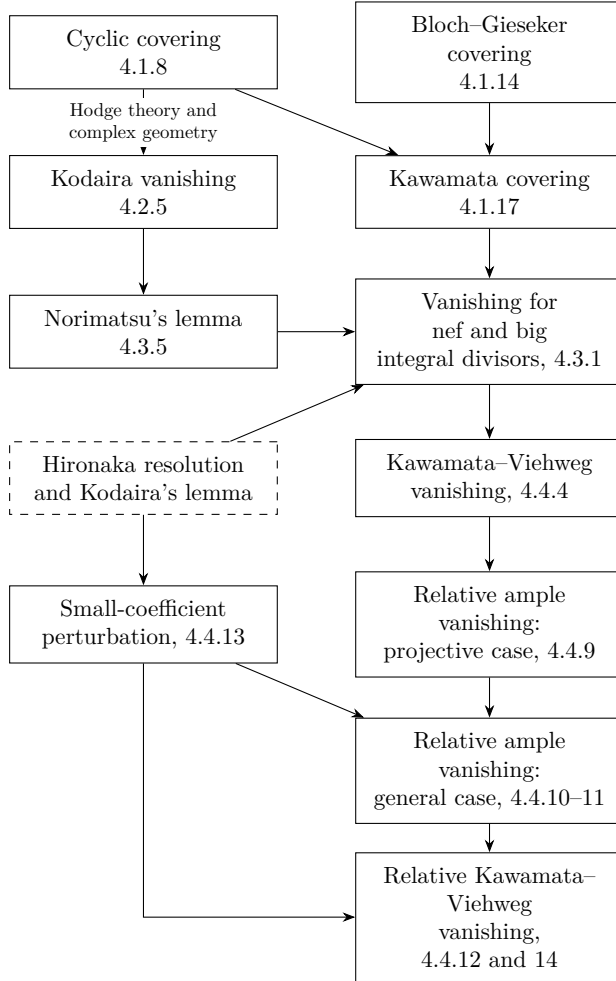


Figure 4.5: Knowledge framework. The perturbation lemma is proved without vanishing and is used in 4.4.10; there is no arrow from relative vanishing back to that lemma. Cyclic and BG coverings also enter the proof of 4.4.4.

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